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**Data:** 10/05/2026

**Título:** Desenvolvimento de Serviços com Spring Boot - TP1

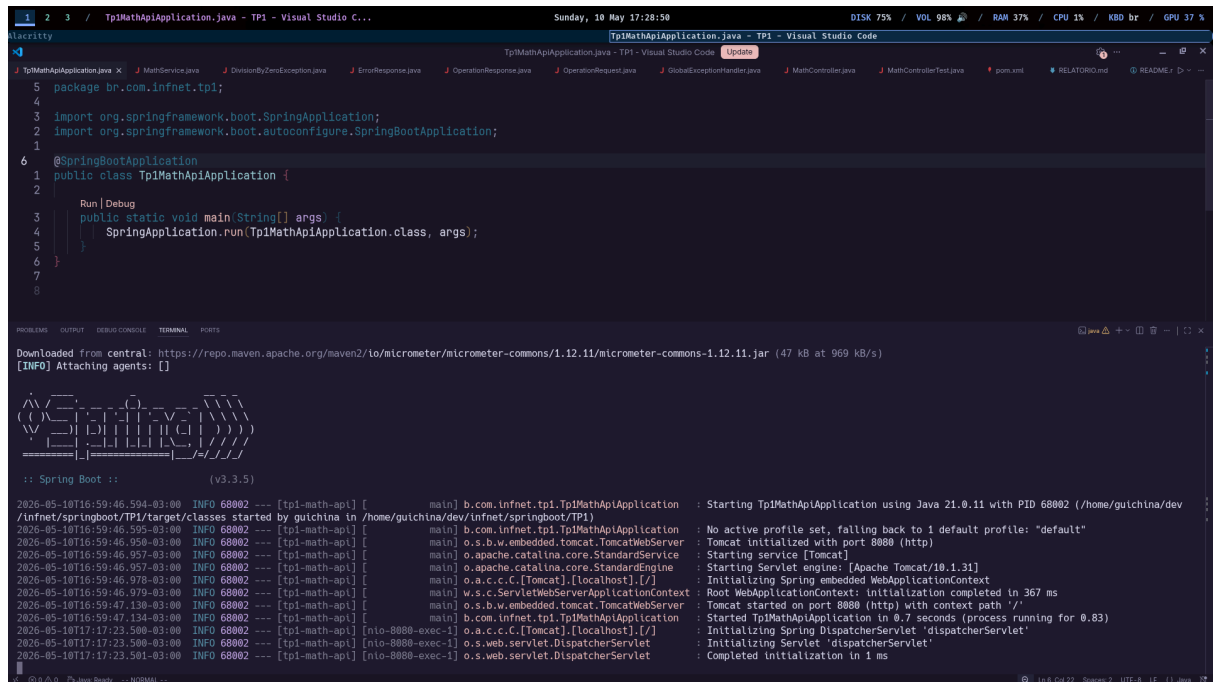
**Link implementação:**

<https://github.com/guilhermecosta/INFNET/tree/master/springboot/TP1>

\* Documentação e relatório do projeto estão na raiz do repositório

**Prints de comprovação:**

**- servidor rodando**



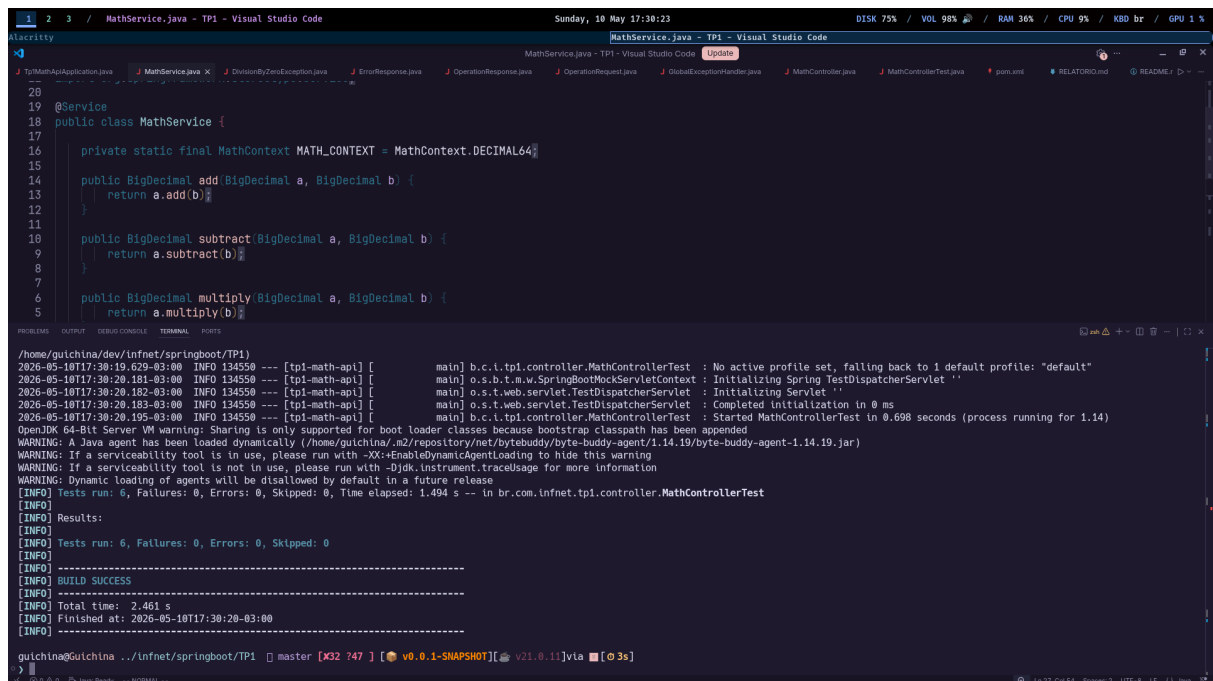
```
1 package br.com.infnet.tp1;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class Tp1MathApiApplication {
8     public static void main(String[] args) {
9         SpringApplication.run(Tp1MathApiApplication.class, args);
10    }
11 }
12
```

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/micrometer/micrometer-commons/1.12.11/micrometer-commons-1.12.11.jar (47 kB at 969 kB/s)
[INFO] Attaching agents: []

:: Spring Boot ::
(v3.3.5)

2026-05-10T16:59:46.594-03:00 INFO 68002 --- [tp1-math-api] [main] b.com.infnet.tp1.Tp1MathApiApplication : Starting Tp1MathApiApplication using Java 21.0.11 with PID 68002 (/home/guichina/dev/infnet/springboot/TP1)
2026-05-10T16:59:46.625-03:00 INFO 68002 --- [tp1-math-api] [main] b.com.infnet.tp1.Tp1MathApiApplication : No active profile set, falling back to 1 default profile: "default"
2026-05-10T16:59:46.950-03:00 INFO 68002 --- [tp1-math-api] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-05-10T16:59:46.957-03:00 INFO 68002 --- [tp1-math-api] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-05-10T16:59:46.957-03:00 INFO 68002 --- [tp1-math-api] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.31]
2026-05-10T16:59:46.978-03:00 INFO 68002 --- [tp1-math-api] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-05-10T16:59:46.979-03:00 INFO 68002 --- [tp1-math-api] [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 367 ms
2026-05-10T16:59:47.138-03:00 INFO 68002 --- [tp1-math-api] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-05-10T16:59:47.134-03:00 INFO 68002 --- [tp1-math-api] [main] b.com.infnet.tp1.Tp1MathApiApplication : Started Tp1MathApiApplication in 0.7 seconds (process running for 0.83)
2026-05-10T17:17:23.508-03:00 INFO 68002 --- [tp1-math-api] [ntio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-05-10T17:17:23.508-03:00 INFO 68002 --- [tp1-math-api] [ntio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-05-10T17:17:23.501-03:00 INFO 68002 --- [tp1-math-api] [ntio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
```

**- testes unitários**



```
20
19 @Service
18 public class MathService {
17
16     private static final MathContext MATH_CONTEXT = MathContext.DECIMAL64;
15
14     public BigDecimal add(BigDecimal a, BigDecimal b) {
13         return a.add(b);
12     }
11
10     public BigDecimal subtract(BigDecimal a, BigDecimal b) {
9         return a.subtract(b);
8     }
7
6     public BigDecimal multiply(BigDecimal a, BigDecimal b) {
5         return a.multiply(b);
4     }
3
2
1

```

```
/home/guichina/dev/infnet/springboot/TP1
2026-05-10T17:30:19.629-03:00 INFO 134558 --- [tp1-math-api] [main] b.c.l.tp1.controller.MathControllerTest : No active profile set, falling back to 1 default profile: "default"
2026-05-10T17:30:20.181-03:00 INFO 134558 --- [tp1-math-api] [main] o.s.b.t.m.w.SpringBootMockServletContext : Initializing Spring TestDispatcherServlet ''
2026-05-10T17:30:20.182-03:00 INFO 134558 --- [tp1-math-api] [main] o.s.t.web.servlet.TestDispatcherServlet : Initializing Servlet ''
2026-05-10T17:30:20.183-03:00 INFO 134558 --- [tp1-math-api] [main] o.s.t.web.servlet.TestDispatcherServlet : Completed initialization in 0 ms
2026-05-10T17:30:20.195-03:00 INFO 134558 --- [tp1-math-api] [main] b.c.l.tp1.controller.MathControllerTest : Started MathControllerTest in 0.698 seconds (process running for 1.14)
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
WARNING: A Java agent has been loaded dynamically (/home/guichina/.m2/repository/net/bytebuddy/byte-buddy-agent/1.14.19/byte-buddy-agent-1.14.19.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.494 s -- in br.com.infnet.tp1.controller.MathControllerTest
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.461 s
[INFO] Finished at: 2026-05-10T17:30:20-03:00
[INFO]
guichina@guichina ~/infnet/springboot/TP1 [master] [X32 747] [v0.0.1-SNAPSHOT] [v21.0.11] [03s]
```

## - Exemplos da api funcionando

```
1 2 3 / Alacritty Sunday, 10 May 17:32:46 DISK 75% / VOL 98% / RAM 38% / CPU 3% / KBD
Alacritty RELATORIO.md - TP1 - Visual Studio Code Alacritty

guichina@Guichina ../infnet/springboot/TP1 % master [x32 ?47 ] [v0.0.1-SNAPSHOT][v21.0.11]via
> curl "http://localhost:8080/api/math/add?a=10&b=5" | jq
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 49 0 49 0 0 21816 0
{
  "operation": "addition",
  "a": 10,
  "b": 5,
  "result": 15
}
```

```
1 2 3 / Alacritty Sunday, 10 May 17:33:09 DISK 75% / VOL 98% / RAM 38% / CPU 3% / KBD
Alacritty RELATORIO.md - TP1 - Visual Studio Code Alacritty

guichina@Guichina ../infnet/springboot/TP1 % master [x32 ?47 ] [v0.0.1-SNAPSHOT][v21.0.11]via
> curl -X POST "http://localhost:8080/api/math/divide" \
-H "Content-Type: application/json" \
-d '{"a":20,"b":4}' | jq
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 62 0 48 100 14 21304 6213
{
  "operation": "division",
  "a": 20,
  "b": 4,
  "result": 5
}
```

```
1 2 3 / Alacritty Sunday, 10 May 17:33:25 DISK 75% / VOL 98% / RAM 38% / CPU 2% / KBD br / GPU 0 %
Alacritty RELATORIO.md - TP1 - Visual Studio Code Alacritty

guichina@Guichina ../infnet/springboot/TP1 % master [x32 ?47 ] [v0.0.1-SNAPSHOT][v21.0.11]via
> curl "http://localhost:8080/api/math/divide?a=10&b=0" | jq
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 160 0 160 0 0 57595 0
{
  "timestamp": "2026-05-10T17:33:24.862172819-03:00",
  "status": 400,
  "error": "Bad Request",
  "message": "Não é permitido dividir por zero.",
  "path": "/api/math/divide"
}
```